EduBandhan — Master Production Deployment Guide

Domain: www.edubandhan.in | **Strategy:** Unified Single-Pod Monorepo | **Host:** Railway

1. Executive Summary & Architecture Overview

EduBandhan adopts a **Unified Single-Container Monorepo Architecture** (Option 1). Both the React 19 Frontend and the FastAPI Backend reside within the same codebase and are built into a high-performance multi-stage Docker container deployed on **Railway**. The PostgreSQL database with the pgvector extension remains cloud-managed on **Neon** (AWS us-east-2).

• **Zero CORS Friction** Frontend uses relative paths (/api/...) because both frontend assets and API endpoints originate from the exact same domain.	• **24/7 WebSockets** Railway persistent container keeps live chat channels and kiosk attendance feeds active without serverless disconnects.	• **Single SSL & Domain** One domain (www.edubandhan.in) serves all dashboards, APIs, and kiosk HUDs under one TLS certificate.

2. Visual Deployment Dataflow

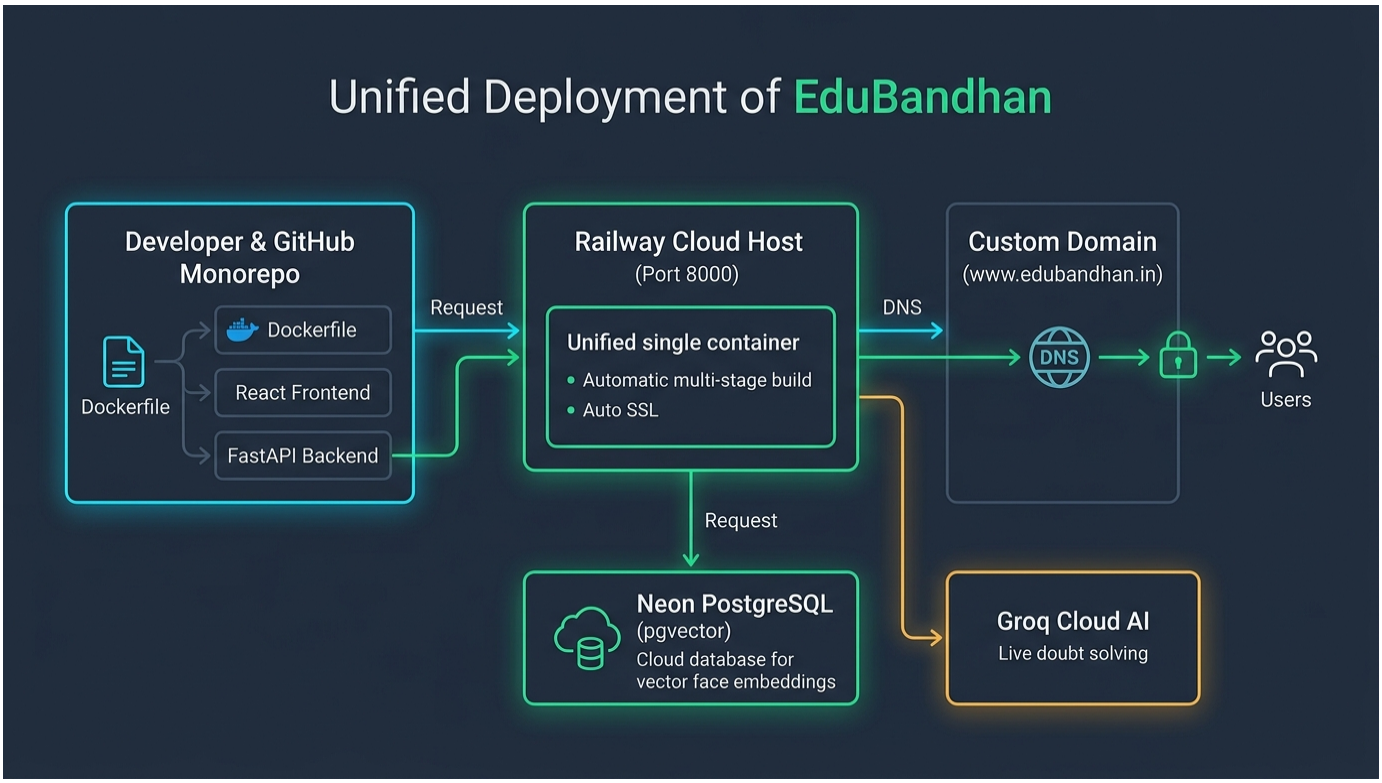


Figure 1: End-to-End Traffic Routing from Domain Registrar through Railway Container to Neon Cloud & Groq AI.

3. Production Environment Secrets Matrix

Copy and paste these exact environment variables into your **Railway Project Dashboard > Service > Variables > Raw Editor**:

Variable Key	Recommended Production Value	Description & Impact
PORT	8000	Unified server listening port
DATABASE_URL	postgresql://neondb_owner:***@ep-***-pooler.c-7.us-east-2.aws.neon.tech/neondb?sslmode=require	Neon pooled connection URL for high-concurrency requests
DIRECT_URL	postgresql://neondb_owner:***@ep-***.c-7.us-east-2.aws.neon.tech/neondb?sslmode=require	Direct non-pooled connection URL for DDL/schema migrations
APP_ENV	production	Production runtime mode (disables debug leak)
JWT_SECRET	edubandhan_super_secret_jwt_key_2026_change_in_production	Cryptographic secret for signing auth session tokens
ALGORITHM	HS256	JWT token hashing algorithm
ACCESS_TOKEN_EXPIRE_MINUTES	43200	Session expiry duration (43,200 min = 30 days)
PLATFORM_FEE_PERCENTAGE	20	School SaaS billing transaction fee percentage
FACE_SIMILARITY_THRESHOLD	0.85	pgvector cosine similarity cutoff for kiosk facial match
CUTOFF_TIME	08:30:00	Morning school attendance cutoff timestamp
GROQ_API_KEY	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	API key for AI Doubt Solver & Parent Teaching Guides
GROQ_MODEL	qwen/qwen3.8-27b	Low-latency instruction-tuned model for school AI

4. Step-by-Step Railway Deployment

Step 1: Push Code to GitHub

Ensure your local repository has the root Dockerfile and .dockerignore. From PowerShell run:
git add . && git commit -m 'Production ready' && git push origin main

Step 2: Initialize Railway Project

Log in to **railway.com** using GitHub. Click **+ New Project** > **Deploy from GitHub repo** > select edubandhan. Railway automatically detects your root multi-stage Dockerfile.

Step 3: Inject Environment Variables

Navigate to your service box > **Variables** tab > click **Raw Editor** and paste the 12 production variables from Section 3. Click **Save**.

Step 4: Automatic Multi-Stage Container Build

Railway builds the React frontend with Node 20, copies dist/ into Python 3.11, starts FastAPI via run.py, and exposes port 8000. Build takes ~2-3 minutes.

5. Custom Domain Configuration (edubandhan.in)

To link your purchased domain **edubandhan.in** to Railway with automatic SSL certification:

1. In Railway > **Settings > Networking > Custom Domains**, add both edubandhan.in and www.edubandhan.in.
2. In your Domain Registrar (Hostinger / GoDaddy / Namecheap / Cloudflare), open **DNS Zone Editor** and add the following records:

Type	Name / Host	Value / Target / RDATA	TTL
CNAME	www	<assigned-app>.up.railway.app	300 (or Auto)
ALIAS / A	@ (root)	<assigned-app>.up.railway.app (or Railway IP)	300 (or Auto)

6. Post-Deployment Verification Checklist

- **Health Check Endpoint:** Navigate to https://www.edubandhan.in/api/health. You should receive HTTP 200 with {"status":"healthy","database":"Neon PostgreSQL (Connected Live)","pgvector_version":"0.8.0"}.
- **Web Portal & Navigation:** Open https://www.edubandhan.in. Ensure all modules (Organization Hierarchy, Teacher Portal, Parent Hub, School Admin) load cleanly without 404s.
- **Biometric Kiosk Attendance:** Open https://www.edubandhan.in/kiosk, scan a test face, and verify real-time cosine similarity matching against pgvector.
- **Real-time WebSockets:** Verify that announcements and live parent-teacher chat connect over wss://www.edubandhan.in/ws/channel/... without dropping.

Status: PRODUCTION READY | Architecture: Option 1 Monorepo | Release: v1.0.0